Structural Inequality and Uneven Labor Market Dynamics: Evidence from Alabama Counties

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Abstract**—***This study examines structural heterogeneity in labor market conditions across counties in Alabama and investigates how these differences shape responses to economic shocks, particularly during the COVID-19 pandemic. Using county-level data from the Local Area Unemployment Statistics (LAUS), the American Community Survey (ACS), and County Business Patterns (CBP), the study constructs a comprehensive dataset capturing both socioeconomic and industrial characteristics.*

A clustering approach based on K-means is employed to classify counties into three distinct groups: Stable, Distressed, and Industrial. The results reveal significant variation in unemployment, income, and poverty levels across these clusters, indicating the presence of substantial structural inequality within the state.

The analysis further shows that the impact of COVID-19 was not uniform across regions. Distressed counties experienced larger increases in unemployment and slower recovery, while Stable counties exhibited greater resilience. Industrial counties demonstrated moderate responses, reflecting their structural composition.

To assess the explanatory power of structural variables, predictive models are developed using Random Forest and XGBoost algorithms. While these models capture baseline differences in unemployment rates, they perform less effectively for high-unemployment regions, suggesting limitations in explaining labor market dynamics using observable factors alone.

Overall, the findings highlight that labor market outcomes within a single state are shaped by underlying structural differences and that these differences play a critical role in determining vulnerability and recovery during economic shocks. The study underscores the importance of region-specific policy approaches in addressing labor market inequality.

I. INTRODUCTION

1.1 Research Background

Economic conditions within a single state are often assumed to be relatively homogeneous. However, substantial variation exists across local labor markets, particularly at the county level. Differences in income levels, poverty rates, and industrial composition create diverse economic environments, which can lead to unequal labor market outcomes even within the same geographic region.

Prior research has consistently emphasized that regional economic disparities are not random but are deeply rooted in structural differences such as industrial specialization, demographic composition, and institutional capacity (Moretti, 2012). These structural factors shape not only long-term economic performance but also the resilience of local labor markets when exposed to external shocks.

The COVID-19 pandemic exposed and amplified these underlying disparities. While some regions experienced temporary disruptions and rapid recovery, others faced prolonged increases in unemployment and slower economic stabilization. Recent studies suggest that the economic impact of COVID-19 was highly uneven across regions, with local characteristics such as industry mix and workforce composition playing a critical role in determining vulnerability and recovery trajectories (Chetty et al., 2020). These differences suggest that pre-existing structural conditions play a critical role in shaping how local labor markets respond to external shocks.

Understanding these regional differences is particularly important for states such as Alabama, where economic conditions vary widely across counties. Despite being treated as a single economic unit in many policy discussions, Alabama contains counties with significantly different socioeconomic and industrial characteristics.

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1.2 Research Problem Statement

Despite growing recognition of regional inequality, there remains a lack of systematic analysis that links structural differences in local economies to labor market outcomes, particularly in the context of economic shocks. Existing approaches often rely on aggregate-level analysis, which masks important variation across counties.

Moreover, while the impact of COVID-19 on unemployment has been widely studied, less attention has been given to how this impact differs across structurally distinct regions within a single state. As a result, it is unclear to what extent structural characteristics can explain differences in vulnerability and recovery, and where their explanatory power becomes limited.

1.3 Research Objectives and Significance

This study aims to identify structural heterogeneity in labor market conditions across Alabama counties and to examine how these differences influence vulnerability and recovery during the COVID-19 shock.

Specifically, the study seeks to:

(1) Classify counties into distinct groups based on socioeconomic and industrial characteristics using clustering techniques,

(2) Analyze how these groups differ in unemployment dynamics during and after the COVID-19 shock, and

(3) Evaluate the extent to which structural variables explain variations in unemployment outcomes.

The significance of this study lies in its ability to provide a more nuanced understanding of regional labor market inequality. By highlighting differences across counties, the findings offer important implications for policymakers, suggesting that uniform policy approaches may be insufficient and that targeted, region-specific interventions are necessary.

II. BACKGROUND

2.1 Structural Determinants of Labor Market Outcomes

The labor market is influenced by a combination of structural and cyclical factors. Structural factors—such as income levels, poverty rates, and industrial composition—determine the baseline conditions of local economies. Counties with higher income levels and diversified industries tend to exhibit more stable employment patterns, while those with higher poverty rates and less diversified economic structures are often more vulnerable to economic downturns.

Prior research highlights that industrial composition plays a central role in shaping local labor market outcomes, as regions specialized in certain sectors are more exposed to structural shocks and long-term employment shifts (Autor & Dorn, 2013).

These structural differences create heterogeneous labor market conditions even within the same state. As a result, regions do not start from the same economic baseline, which leads to variation in unemployment levels and long-term labor market stability.

2.2 Regional Disparities and Labor Market Inequality

Previous research has shown that regional disparities in labor market outcomes are closely linked to structural characteristics. Areas with concentrated economic disadvantage often experience persistently higher unemployment rates and slower recovery following economic shocks. In contrast, economically stronger regions tend to maintain lower unemployment and recover more quickly.

Industrial composition also plays a key role in shaping these disparities. Regions that are heavily dependent on specific industries may be more exposed to sector-specific risks, while more diversified economies tend to be more resilient. These patterns highlight the presence of structural inequality across regions, even within a single state.

2.3 COVID-19 as an Exogenous Shock

The COVID-19 pandemic provides a unique context to examine these dynamics. As a large and sudden external shock, it disrupted labor markets across all regions but did not affect them uniformly. Differences in industry exposure, workforce characteristics, and local economic resilience contributed to varying levels of impact and recovery.

Recent evidence shows that the economic effects of COVID-19 varied significantly across local economies, particularly affecting small businesses and regions with limited financial resilience (Bartik et al., 2020).

While prior studies have examined the overall effect of COVID-19 on employment, fewer have focused on within-state variation at the county level. This limitation makes it difficult to fully understand how structural differences influence vulnerability and recovery across local labor markets.

2.4 Data-Driven Approaches to Labor Market Analysis

Recent advances in data science and machine learning have enabled more granular analysis of labor market dynamics using large-scale datasets. Unlike traditional approaches that rely on aggregated statistics, data-driven methods allow for the identification of hidden patterns and structural heterogeneity across regions.

Clustering techniques, in particular, provide a systematic way to group regions with similar socioeconomic and industrial characteristics, enabling a more nuanced understanding of regional disparities. In addition, predictive modeling can be used to assess the extent to which observable structural variables explain labor market outcomes, while also revealing the limitations of such explanations.

By integrating data-driven methods with economic analysis, this study leverages computational techniques to uncover latent structures in regional labor markets and to evaluate how these structures relate to differential responses to economic shocks.

III. METHDOLOGY

3.1 Data Sources and Integration

This study utilizes multiple publicly available datasets to construct a county-level dataset for the state of Alabama. Monthly unemployment data are obtained from the Local Area Unemployment Statistics (LAUS), while socioeconomic indicators such as population, median income, and poverty rate are collected from the American Community Survey (ACS). In addition, industrial structure variables are derived from the County Business Patterns (CBP), which provide information on the number of establishments across different sectors.

The datasets are merged using county-level Federal Information Processing Standards (FIPS) codes to ensure consistency across sources. Temporal alignment is applied to match variables across different frequencies, resulting in a unified dataset covering the period from 2015 to 2024. Monthly unemployment rates are retained to preserve temporal variation and capture short-term fluctuations in labor market conditions. The integration of multiple data sources enables a comprehensive analysis of both structural and dynamic aspects of local labor markets. The datasets are merged using county-level Federal Information Processing Standards (FIPS) codes and aligned temporally to create a unified dataset spanning from 2015 to 2024. Monthly unemployment rates are aggregated to annual averages when necessary to ensure consistency across datasets.

3.2 Data Preprocessing and Feature Construction

The dataset undergoes several preprocessing steps to ensure consistency and accuracy. First, only counties within Alabama are retained by filtering observations based on state-level FIPS codes. County identifiers are standardized to enable accurate merging across datasets.

Key variables are constructed to capture both labor market outcomes and structural characteristics. The unemployment rate is derived directly from LAUS data at the monthly level. Socioeconomic variables, including population, median income, and poverty rate, are obtained from ACS and merged at the county-year level. Industrial structure variables are computed using CBP data, where sectoral shares—such as manufacturing, retail, and healthcare—are calculated as the proportion of establishments in each sector relative to the total number of establishments within each county.

To illustrate overall labor market trends, Figure 1 presents the average monthly unemployment rate across Alabama counties over time. As you can check in Figure 1, the unemployment rate shows a sharp increase in 2020, corresponding to the COVID-19 shock, followed by a gradual decline in subsequent years.

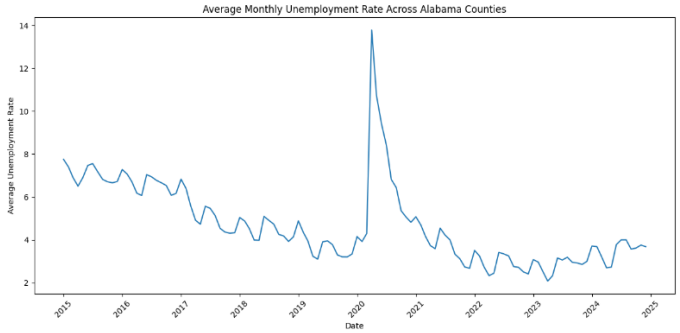

Figure 1. Average Monthly Unemployment Rate

3.3 Clustering Analysis

To identify structural heterogeneity across counties, a clustering approach is applied using the K-means algorithm. County-level features—including unemployment rate, poverty rate, median income, and industrial shares—are first standardized to ensure comparability across variables with different scales.

The K-means algorithm is then used to partition counties into three clusters, representing distinct labor market structures. The resulting clusters are interpreted as follows:

- **Stable (Cluster 0)**: low unemployment, higher income
- **Distressed (Cluster 1)**: high unemployment, higher poverty
- **Industrial (Cluster 2)**: moderate unemployment with higher manufacturing presence

To further examine the characteristics of each cluster, summary statistics are calculated based on the clustering results. Table 1 presents the average values of key variables within each cluster, including unemployment rate, poverty rate, median income, and industrial composition. These statistics provide a quantitative basis for interpreting the differences across clusters.

Cluster	Unemp.	Poverty	Income ($k)	Mfg.	Retail	Health
Stable (0)	3.81	0.15	58.2	0.04	0.17	0.10
Distressed (1)	6.43	0.27	33.9	0.05	0.23	0.09
Industrial (2)	4.33	0.19	44.9	0.06	0.20	0.10

Table 1. Cluster Summary Statistics

Note
Unemp. = Unemployment rate; Mfg. = Manufacturing; Med. Income = Median household income.

As shown in Table 1, the "Distressed" cluster exhibits the highest average unemployment rate and poverty level, along with the lowest median income, indicating structurally disadvantaged economic conditions. In contrast, the "Stable" cluster demonstrates lower unemployment and poverty rates combined with higher income levels, suggesting stronger and more resilient local economies. The "Industrial" cluster lies between these two groups in terms of socioeconomic conditions but is distinguished by a relatively higher share of manufacturing activity, reflecting its industrial composition.

These results confirm that the clusters capture meaningful structural differences rather than arbitrary groupings. The variation in both socioeconomic indicators and industry structure suggests that counties in Alabama operate under significantly different economic conditions, even within the same state.

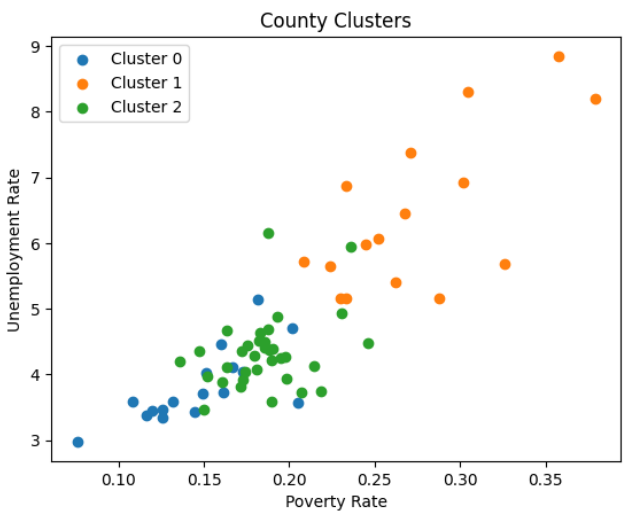

Figure 2. Cluster Visualization

Figure 2 provides a visual representation of the clustering results using a scatter plot of counties based on selected features. As you can check in Figure 2, the clusters form clearly distinguishable groups, indicating effective separation of counties according to their structural characteristics. The visual clustering supports the interpretation derived from the summary statistics.

To further validate the clustering results, an analysis of variance (ANOVA) test is conducted to assess whether the differences in unemployment rates across clusters are statistically significant. The results show a statistically significant difference between groups, confirming that the clustering captures real variation in labor market outcomes rather than random noise.

Overall, the clustering analysis provides a data-driven classification of counties that reflects underlying structural differences in local economies. This classification serves as a foundation for subsequent analysis, particularly in examining how different types of counties respond to external shocks such as the COVID-19 pandemic.

3.4 COVID-19 Shock and Recovery Analysis

To examine the impact of the COVID-19 pandemic on local labor markets, this study compares unemployment rates across three distinct periods: the pre-shock period (before 2020), the shock period (2020), and the recovery period (after 2022).

For each cluster, key metrics are calculated to quantify the magnitude of the shock, the peak unemployment level, and the recovery pattern. This approach enables a comparative analysis of how structurally different regions respond to external economic shocks.

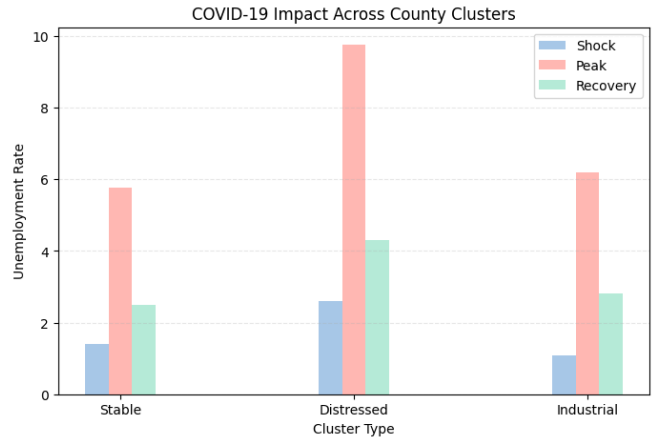

Figure 3. COVID-19 Impact Across County Clusters

Figure 3 presents the differences in shock magnitude, peak unemployment, and recovery across clusters. As you can check in Figure 3, the "Distressed" cluster experienced the largest increase in unemployment during the shock period and maintained higher unemployment levels during recovery. In contrast, the "Stable" cluster shows relatively smaller disruptions and faster recovery, while the "Industrial" cluster demonstrates moderate impact and resilience.

3.5 Predictive Analysis (Supporting Evidence)

To further evaluate the explanatory power of structural variables, predictive models are developed using Random Forest and XGBoost algorithms. These models use socioeconomic and industrial features to predict county-level unemployment rates. The purpose of this analysis is not to establish a definitive forecasting model, but rather to assess how much variation in unemployment can be explained by observable structural characteristics.

Table 2 summarizes the performance of the predictive models using evaluation metrics such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). The results indicate moderate predictive performance, suggesting that structural variables capture baseline variation in unemployment but do not fully explain all fluctuations.

Model	MAE	RMSE
Random Forest	1.30	1.49
XGBoost	1.25	1.47

Table 2. Model Performance (MAE and RMSE)

Figure 4 presents the feature importance results from the predictive models, showing the relative contribution of each variable. Median income and poverty rate are among the most influential predictors, indicating the importance of structural socioeconomic factors in shaping labor market outcomes.

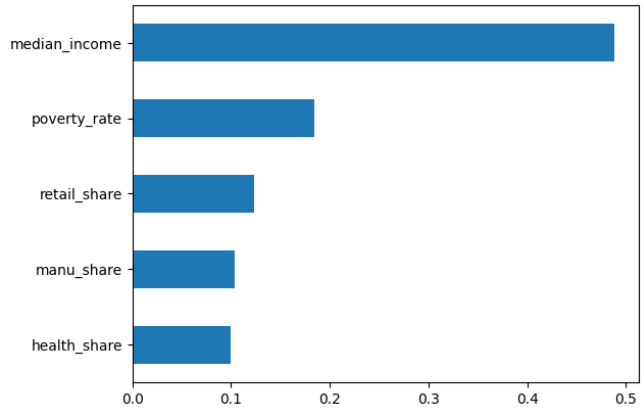

Figure 4. Feature Importance

However, prediction performance is not uniform across all observations. The model tends to perform relatively well for lower unemployment levels but exhibits larger errors for higher unemployment counties. This pattern suggests that structurally disadvantaged regions are more difficult to predict using observable variables alone.

This finding indicates that while structural variables explain baseline labor market conditions, they are insufficient to fully capture the dynamics of economic shocks and recovery. This limitation highlights the role of additional unobserved factors, such as policy responses and localized economic conditions, in shaping labor market outcomes.

IV. Result

4.1 Cluster Characteristics

The clustering analysis reveals clear structural differences across counties in Alabama. The three identified clusters—Stable, Distressed, and Industrial—exhibit distinct socioeconomic and labor market characteristics.

The "Stable" cluster is characterized by relatively low unemployment rates and higher median income levels, indicating stronger economic conditions. In contrast, the "Distressed" cluster exhibits significantly higher unemployment and poverty rates, along with lower income levels, reflecting structural economic disadvantage. The "Industrial" cluster falls between these two groups, with moderate unemployment levels and a relatively higher share of manufacturing activity.

These differences suggest that counties in Alabama do not share a uniform economic structure but instead represent heterogeneous labor market environments. This structural heterogeneity forms the basis for understanding differences in how counties respond to economic shocks. The spatial distribution of clusters is provided in Appendix A (Figure A1) for additional reference.

4.2 Differential Impact of COVID-19

The impact of the COVID-19 pandemic varies significantly across clusters. To evaluate this effect, changes in unemployment rates are analyzed across the pre-shock, shock, and recovery periods.

Figure 3 illustrates the magnitude of the shock, peak unemployment levels, and recovery patterns across clusters. As you can check in Figure 3, the "Distressed" cluster experienced the largest increase in unemployment during the shock period. This cluster also reached the highest peak unemployment levels and exhibited slower recovery compared to other clusters.

In contrast, the "Stable" cluster experienced a smaller increase in unemployment and returned more quickly to pre-pandemic levels. The "Industrial" cluster shows moderate impact, suggesting partial resilience likely linked to its industrial composition.

These findings indicate that pre-existing structural conditions play a critical role in shaping the magnitude of economic shocks and the speed of recovery.

4.3 Spatial Distribution of Clusters

The spatial distribution of clusters provides additional insight into regional patterns of economic inequality. Figure 4 presents a geographic visualization of cluster assignments across Alabama counties.

As you can check in Figure 4, counties belonging to the "Distressed" cluster are geographically concentrated in specific regions, while "Stable" counties are more widely distributed. This spatial clustering suggests that economic disadvantage is not randomly distributed but instead follows regional patterns.

This geographic concentration reinforces the idea that regional inequality is structurally embedded and highlights the importance of location-specific factors in shaping labor market outcomes.

4.4 Predictive Analysis and Model Limitations

To further assess the explanatory power of structural variables, predictive models are developed to estimate unemployment rates using socioeconomic and industrial features.

The results show that the models achieve moderate predictive performance, indicating that structural variables capture baseline differences in labor market conditions. However, prediction errors are not uniform across all observations.

The model performs relatively well for counties with lower unemployment rates but exhibits larger errors for counties with higher unemployment. This pattern suggests that structurally disadvantaged regions are more difficult to predict using observable variables alone.

This finding implies that while structural characteristics explain general patterns, they are insufficient to fully capture the complexity of labor market dynamics, particularly under conditions of economic stress. Additional factors—such as policy responses, firm behavior, and localized economic conditions—likely play an important role.

V. CONCLUSION

5.1 Conclusion

This study examines structural heterogeneity in labor market conditions across counties in Alabama and evaluates how these differences shape responses to economic shocks, particularly during the COVID-19 pandemic.

Using a clustering approach, the analysis identifies three distinct groups of counties—Stable, Distressed, and Industrial—each characterized by different socioeconomic and industrial structures. The results show that these structural differences are strongly associated with variations in unemployment outcomes.

The findings further demonstrate that the impact of COVID-19 was not uniform across regions. Counties classified as “Distressed” experienced significantly larger increases in unemployment and slower recovery, while “Stable” counties showed relatively mild disruptions and faster recovery. These results highlight the importance of pre-existing structural conditions in determining both vulnerability to economic shocks and the speed of recovery.

In addition, the predictive analysis reveals that while structural variables explain baseline differences in unemployment, they are insufficient to fully capture labor market dynamics, particularly in high-unemployment regions. This suggests that additional factors beyond observable structural characteristics play a critical role in shaping economic outcomes.

Overall, this study provides evidence that labor market conditions within a single state are far from homogeneous and are instead shaped by underlying structural inequalities.

5.2 Policy Implications

The findings of this study have important implications for economic policy and regional development strategies.

First, the results suggest that uniform, state-wide policy approaches may be insufficient to address labor market disparities. Given the substantial structural differences across counties, policies should be tailored to the specific conditions of each region. In particular, counties classified as “Distressed” may require targeted interventions, such as workforce development programs, income support, and investments in local economic infrastructure. This aligns with prior research emphasizing that effective economic policies must be context-specific rather than uniformly applied across regions (Rodrik, 2004).

Second, the strong association between socioeconomic conditions and labor market outcomes indicates the need for long-term structural policies. Efforts to reduce poverty and increase income levels may have significant effects on improving labor market stability and resilience.

Third, the findings highlight the importance of industrial diversification. Regions that rely heavily on a limited number of industries may be more vulnerable to economic shocks, suggesting that policies promoting diversification could enhance economic resilience.

Finally, the limitations of predictive models indicate that policymakers should exercise caution when relying solely on data-driven forecasts. While such models are useful for identifying general patterns, they may fail to capture complex local dynamics, particularly in economically vulnerable regions. Therefore, quantitative analysis should be complemented with qualitative and region-specific insights.

VI. Citation

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VII. Appendix

Appendix A. Spatial Distribution of County Clusters

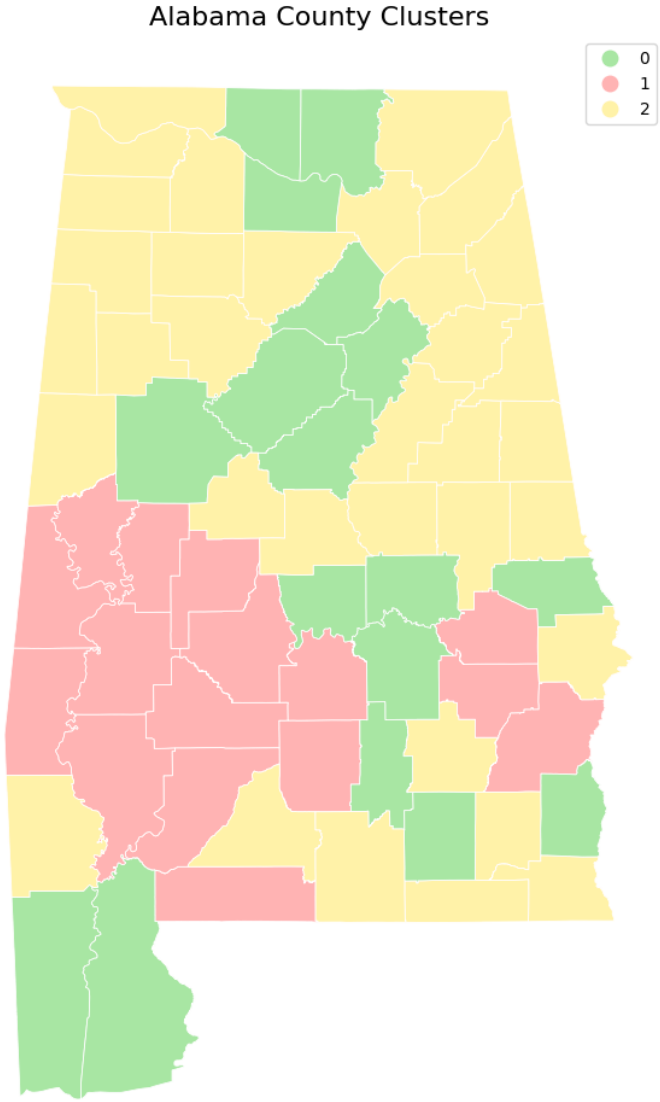

Figure A1. Spatial Distribution of County Clusters in Alabama

- **Stable (Cluster 0)**: low unemployment, higher income
- **Distressed (Cluster 1)**: high unemployment, higher poverty
- **Industrial (Cluster 2)**: moderate unemployment with higher manufacturing presence

This figure presents the geographic distribution of clusters across Alabama counties. Counties are categorized into three groups based on structural labor market characteristics: Stable (green), Distressed (red), and Industrial (yellow). The visualization highlights the regional concentration of economically distressed areas and provides additional spatial context for the clustering results.

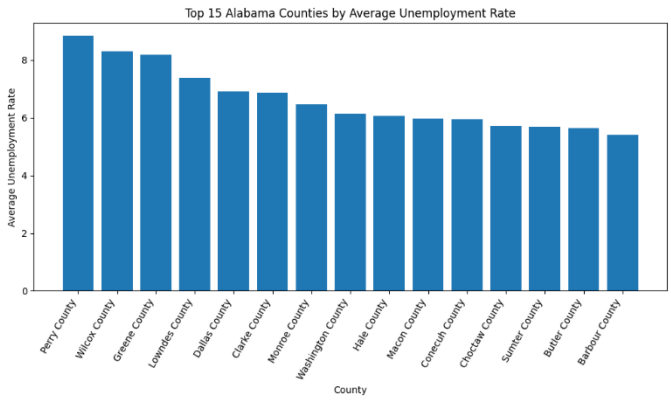

Figure A2. Top 15 Alabama Counties by Average Unemployment Rate B

Figure A2 presents the 15 counties in Alabama with the highest average unemployment rates over the study period. As shown in the figure, these counties consistently exhibit elevated unemployment levels relative to others in the state.

This pattern indicates the presence of persistent structural disadvantages. Counties with high average unemployment are likely associated with weaker economic conditions, including lower income levels and higher poverty rates, reinforcing the clustering results that identify structurally vulnerable regions.

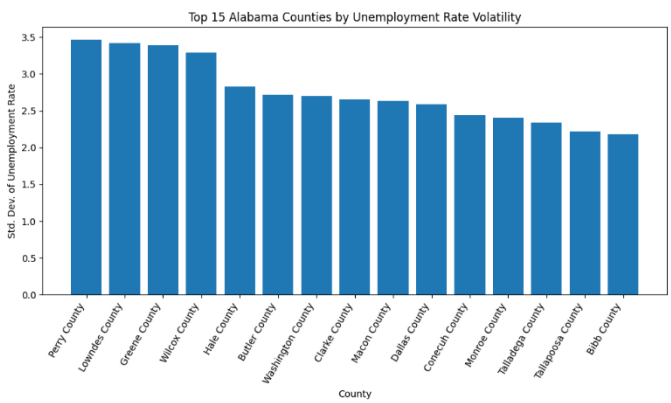

Figure A3. Top 15 Alabama Counties by Unemployment Rate Volatility

Figure A3 displays the 15 counties with the highest unemployment rate volatility. These counties experience larger fluctuations in unemployment over time, indicating higher sensitivity to economic changes and shocks.

Such volatility suggests unstable labor market conditions, where employment levels are more susceptible to external disruptions. This finding supports the idea that certain regions not only face higher unemployment but also experience greater instability.

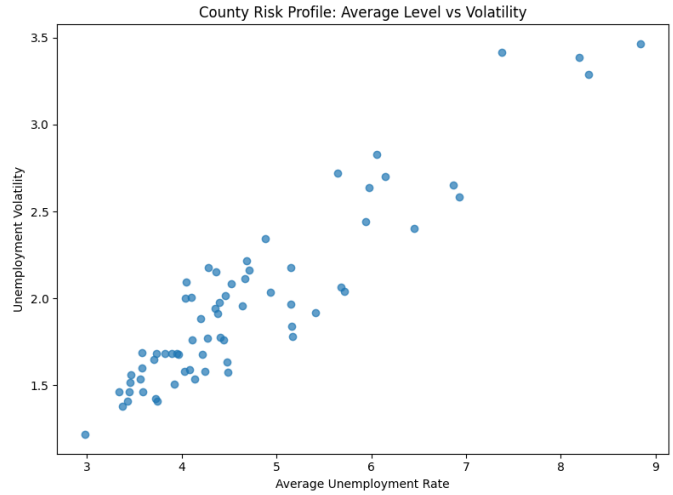

Figure A4. County Risk Profile: Average Level vs Volatility

Figure A4 illustrates the relationship between average unemployment levels and unemployment volatility across counties. As shown in the figure, there is a clear positive relationship between the two variables.

This implies that counties with higher average unemployment also tend to experience greater volatility, indicating compounded economic risk. Counties located in the upper-right region of the plot can be interpreted as the most vulnerable, as they combine both high unemployment levels and high instability.

TableA1. Full Cluster Assignment for each county

county_name	fips	cluster	cluster_label
Autauga County	01001	0	Stable
Baldwin County	01003	0	Stable
Blount County	01009	0	Stable
Coffee County	01031	0	Stable
Crenshaw County	01041	0	Stable
Elmore County	01051	0	Stable
Henry County	01067	0	Stable
Jefferson County	01073	0	Stable
Lee County	01081	0	Stable
Limestone County	01083	0	Stable
Madison County	01089	0	Stable
Mobile County	01097	0	Stable
Montgomery County	01101	0	Stable
Morgan County	01103	0	Stable
Shelby County	01117	0	Stable
St. Clair County	01115	0	Stable
Tuscaloosa County	01125	0	Stable
Barbour County	01005	1	Distressed
Bullock County	01011	1	Distressed
Butler County	01013	1	Distressed
Choctaw County	01023	1	Distressed
Clarke County	01025	1	Distressed
Dallas County	01047	1	Distressed
Escambia County	01053	1	Distressed
Greene County	01063	1	Distressed
Hale County	01065	1	Distressed
Lowndes County	01085	1	Distressed
Macon County	01087	1	Distressed
Marengo County	01091	1	Distressed
Monroe County	01099	1	Distressed
Perry County	01105	1	Distressed
Sumter County	01119	1	Distressed
Wilcox County	01131	1	Distressed
Bibb County	01007	2	Industrial
Calhoun County	01015	2	Industrial
Chambers County	01017	2	Industrial
Cherokee County	01019	2	Industrial

Chilton County	01021	2	Industrial
Clay County	01027	2	Industrial
Cleburne County	01029	2	Industrial
Colbert County	01033	2	Industrial
Conecuh County	01035	2	Industrial
Coosa County	01037	2	Industrial
Covington County	01039	2	Industrial
Cullman County	01043	2	Industrial
Dale County	01045	2	Industrial
DeKalb County	01049	2	Industrial
Etowah County	01055	2	Industrial
Fayette County	01057	2	Industrial
Franklin County	01059	2	Industrial
Geneva County	01061	2	Industrial
Houston County	01069	2	Industrial
Jackson County	01071	2	Industrial
Lamar County	01075	2	Industrial
Lauderdale County	01077	2	Industrial
Lawrence County	01079	2	Industrial
Marion County	01093	2	Industrial
Marshall County	01095	2	Industrial
Pickens County	01107	2	Industrial
Pike County	01109	2	Industrial
Randolph County	01111	2	Industrial
Russell County	01113	2	Industrial
Talladega County	01121	2	Industrial
Tallapoosa County	01123	2	Industrial
Walker County	01127	2	Industrial
Washington County	01129	2	Industrial
Winston County	01133	2	Industrial

https://docs.google.com/spreadsheets/d/1hhSBAG3Jx8RU1lmcNo7Zm7a_wY2HCG7wsm_IBBXfFdk/edit?gid=0#gid=0